

Test of Mathematics for University Admission
Mock Set E Paper 1
 Applications of Mathematical Knowledge

Name: _____

Date sat: _____

Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content only. The difficulty lies in how that content is used: parameters that change the case, conditions that quietly remove a solution, and answers that come from structure rather than from brute force.

[illegible]

Question 1

Real numbers x and y satisfy $x + y = 4$ and $x^3 + y^3 = 40$. Find the value of $x^4 + y^4$.

- A 120
- B 128
- C 136
- D 144
- E 152
- F 160

Question 2

Find the product of all the real solutions of

$$2 \log_x 4 - \log_2 x = 1.$$

- A $\frac{1}{4}$
- B $\frac{1}{2}$
- C 1
- D 2
- E 4
- F $\sqrt{2}$

Question 3

The point $(1, 2)$ is reflected in the line $y = 2x - 3$. Find the coordinates of its image.

- A $(\frac{17}{5}, \frac{4}{5})$
- B $(3, 1)$
- C $(\frac{11}{5}, \frac{2}{5})$
- D $(-1, 3)$
- E $(\frac{13}{5}, \frac{6}{5})$
- F $(\frac{17}{5}, -\frac{4}{5})$

Question 4

Evaluate

$$\sum_{k=1}^{48} \frac{1}{\sqrt{k} + \sqrt{k+2}}.$$

- A $2\sqrt{2}$
- B $3 + \sqrt{2}$
- C 6
- D $3 + 2\sqrt{2}$
- E $4 + 2\sqrt{2}$
- F $7 - \sqrt{2}$

Question 5

In the expansion of $(2 + 3x)^{10}$ in ascending powers of x , the coefficient of x^k is greatest when k equals

- A 2
- B 3
- C 4
- D 5
- E 6
- F 7

Question 6

How many real solutions does the equation $2^x = 3 - x^2$ have?

- A 1
- B 2
- C 3
- D 4
- E 5

Question 7

A geometric series has sum to infinity 12, and the series formed by squaring each of its terms has sum to infinity 48. Find the sum to infinity of the series formed by cubing each term.

- A 144
- B 216
- C 288
- D 432
- E $\frac{1728}{5}$
- F $\frac{1728}{7}$

Question 8

An angle x satisfies $0^\circ < x < 90^\circ$ and $\sin x + \cos x = \frac{7}{5}$. Find the sum of all possible values of $\tan x$.

- A $\frac{3}{4}$
- B $\frac{4}{3}$
- C $\frac{25}{12}$
- D $\frac{7}{5}$
- E $\frac{12}{7}$
- F $\frac{24}{7}$

Question 9

Find the total area of the regions enclosed between the curves $y = x^3$ and $y = 4x$.

- A 2
- B 4
- C 6
- D 8
- E 16
- F 32

Question 10

The tangents from $P(7, 1)$ to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ touch the circle at A and B . Find the area of the quadrilateral $PACB$, where C is the centre of the circle.

- A $3\sqrt{7}$
- B $6\sqrt{7}$
- C 9
- D 12
- E $3\sqrt{37}$
- F $\sqrt{37}$

Question 11

Find the complete set of real values of k for which

$$(k - 1)x^2 + 2kx + k + 3 \geq 0 \quad \text{for every real } x.$$

- A $k \geq \frac{3}{2}$
- B $k > \frac{3}{2}$
- C $k > 1$
- D $k \geq 1$
- E $1 < k \leq \frac{3}{2}$
- F $k \leq \frac{3}{2}$

Question 12

Positive real numbers x and y satisfy $x^2y = 32$. Find the minimum value of $x + 2y$.

- A 6
- B $4\sqrt[3]{4}$
- C $6\sqrt[3]{2}$
- D 8
- E $8\sqrt{2}$
- F 12

Question 13

Find the complete set of real values of k for which the equation $\cos^2 x - \sin x = k$ has at least one real solution x .

- A $-1 \leq k \leq 1$
- B $0 \leq k \leq \frac{5}{4}$
- C $-\frac{5}{4} \leq k \leq 1$
- D $-1 \leq k \leq 2$
- E $-1 \leq k \leq \frac{5}{4}$
- F $k \leq \frac{5}{4}$

Question 14

The triangle with vertices $(0, 0)$, $(4, 1)$ and $(k, 5)$ has area 10. Find the sum of all possible values of k .

- A 0
- B 10
- C 20
- D 30
- E 35
- F 40

Question 15

Two circles of radius 1 have their centres a distance 1 apart. Find the area of the region inside both circles.

- A $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$
- B $\frac{\pi}{3}$
- C $\frac{2\pi}{3} - \sqrt{3}$
- D $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$
- E $\frac{\pi}{2} - \frac{\sqrt{3}}{4}$
- F $\frac{4\pi}{3} - \sqrt{3}$

Question 16

The normal to the curve $y = x^2$ at the point $P(1, 1)$ meets the curve again at Q . Find the length PQ .

- A $\frac{5}{2}$
- B $\frac{9}{4}$
- C $\frac{5\sqrt{2}}{2}$
- D $\frac{3\sqrt{5}}{2}$
- E $\frac{5\sqrt{5}}{4}$
- F $\frac{5\sqrt{3}}{2}$

Question 17

For which value of a is

$$\int_a^{a+1} (x^2 - 4x) dx$$

as small as possible?

- A 1
- B $\frac{3}{2}$
- C 2
- D $\frac{5}{2}$
- E 3
- F 0

Question 18

Let $a > 0$. For which values of a is the greatest value of $f(x) = x^3 - 3a^2x$ on the interval $0 \leq x \leq 2$ equal to 0?

- A $a \geq 2$
- B $a \geq 1$
- C $a \geq \sqrt{2}$
- D $0 < a \leq \frac{2}{\sqrt{3}}$
- E $a \geq \frac{4}{3}$
- F $a \geq \frac{2}{\sqrt{3}}$

Question 19

Find the total area of the regions enclosed between the curve $y = \frac{x-2}{\sqrt{x}}$, the x -axis and the lines $x = 1$ and $x = 4$.

- A $\frac{16\sqrt{2} - 18}{3}$
- B $\frac{2}{3}$
- C $\frac{8\sqrt{2} - 10}{3}$
- D $\frac{8\sqrt{2} - 8}{3}$
- E $\frac{16\sqrt{2} - 10}{3}$
- F $\frac{10}{3}$

Question 20

The sequence $u_1, u_2, u_3, \dots$ is defined by $u_{n+1} = u_n^2 - 2u_n + 2$. For which values of u_1 does the sequence remain bounded?

- A $0 < u_1 < 2$
- B $1 \leq u_1 \leq 2$
- C $-1 \leq u_1 \leq 1$
- D $0 \leq u_1 \leq 2$
- E $u_1 \leq 2$
- F every real u_1

END OF PAPER